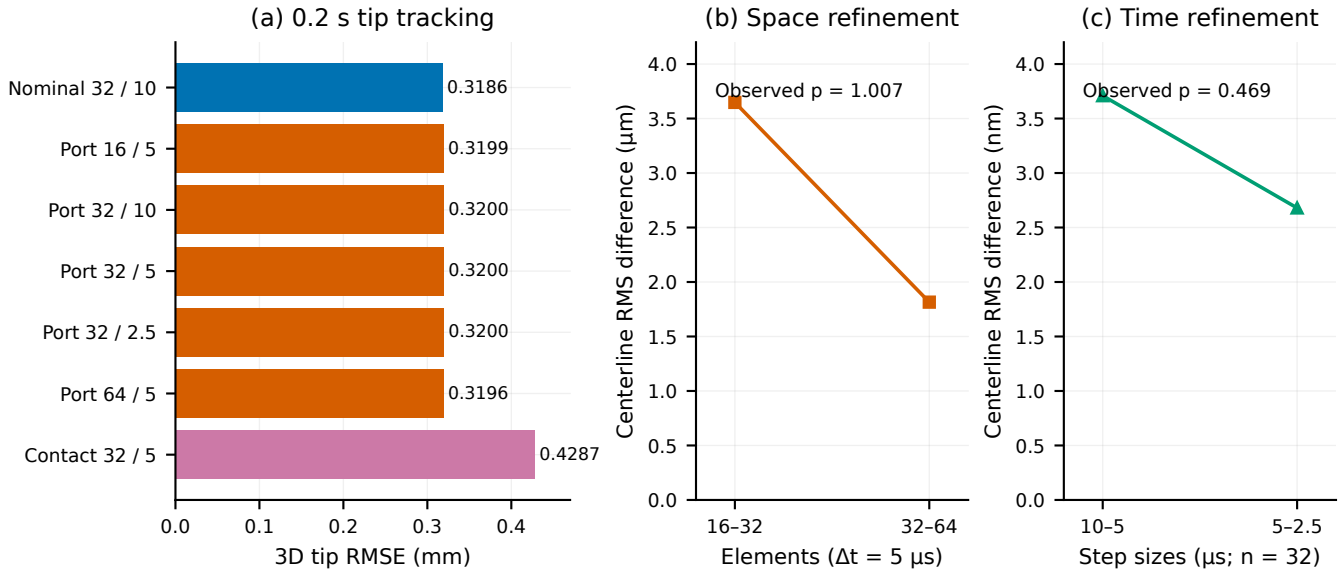
(a) 0.2 s tip tracking
Nominal 32 / 10
Port 16 / 5
Port 32 / 10
Port 32 / 5
Port 32 / 2.5
Port 64 / 5
Contact 32 / 5
0.3186
0.3199
0.3200
0.3200
0.3200
0.3196
0.4287
0.0
0.1
0.2
0.3
0.4
3D tip RMSE (mm)
(b) Space refinement
Observed p = 1.007
Centerline RMS difference (μm)
16–32
32–64
Elements (Δt = 5 μs)
(c) Time refinement
Observed p = 0.469
Centerline RMS difference (nm)
10–5
5–2.5
Step sizes (μs; n = 32)